📊 Sentiment Analysis on “AI Replacing Jobs” Tweets

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1. Introduction

The topic chosen for this project is **“AI replacing jobs”**, which is highly relevant in today’s technological era. With the rapid growth of artificial intelligence and automation, there is increasing discussion about its impact on employment and the future workforce.

This topic was selected because it reflects real-world concerns and diverse public opinions. While some individuals believe that AI will create new opportunities and improve productivity, others fear that it may lead to job loss and economic instability. These mixed opinions make it an ideal subject for sentiment analysis.

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2. Data Collection & Preprocessing

A total of **100 tweets** related to the topic “AI replacing jobs” were collected. The dataset was manually created to simulate realistic opinions similar to those found on social media platforms.

Data Preprocessing Steps:

- Converted all text to lowercase
- Removed punctuation, numbers, and special characters
- Eliminated unnecessary spaces

Data Labeling:

Each tweet was manually classified into one of the following categories:

- **Positive**: Tweets supporting AI or highlighting its benefits
- **Negative**: Tweets expressing concern about job loss due to AI
- **Neutral**: Tweets that are balanced, unclear, or informational

Manual labeling ensures better control over the dataset and improves classification accuracy.

3. Data Splitting

The dataset was divided into:

- **80 tweets (80%) for training**
- **20 tweets (20%) for testing**

This split was chosen because it provides enough data for training the models while keeping a portion separate for unbiased evaluation. The 80-20 split is a standard practice in machine learning.

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4. Model Training & Classification

Three machine learning classifiers were used to perform sentiment analysis:

- **Naïve Bayes**
- **Logistic Regression**
- **Support Vector Machine (SVM)**

Training Process:

- The text data was converted into numerical format using **CountVectorizer**
- The models were trained on the training dataset (80 tweets)
- Predictions were made on the testing dataset (20 tweets)

Hyperparameters Used:

- Logistic Regression: max_iter = 200
- Naïve Bayes: Default settings
- SVM: Default settings

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5. Model Evaluation

The models were evaluated using the following metrics:

- Precision
- Recall
- F1-score
- Accuracy

Results Summary:

Naïve Bayes:

- Accuracy: 75%
- Macro Average Precision: 0.79
- Macro Average Recall: 0.69

Logistic Regression:

- Accuracy: 60%
- Macro Average Precision: 0.64
- Macro Average Recall: 0.51

Support Vector Machine (SVM):

- Accuracy: 60%
- Macro Average Precision: 0.64
- Macro Average Recall: 0.51

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6. Results & Observations

The results indicate that:

- **Naïve Bayes performed the best**, achieving the highest accuracy and better precision and recall compared to other models.
- **Logistic Regression and SVM performed similarly**, but with lower accuracy and weaker recall, especially for neutral and negative sentiments.
- All models showed better performance in predicting **positive sentiments**, while neutral and negative classifications were more challenging.

This is likely due to the small dataset size and the complexity of interpreting neutral or ambiguous text.

Overall, Naïve Bayes proved to be the most suitable model for this dataset due to its effectiveness in handling text classification problems.

7. Screenshots / Results

- Dataset Screenshot

tweet	label
AI is transforming jobs in a positive way	positive
Automation is going to eliminate many roles	negative
Not sure how AI will affect employment	neutral
AI tools are making work faster and smarter	positive
I'm afraid robots will replace humans soon	negative
AI could create more jobs than it destroys	positive
The impact of AI on jobs is still unclear	neutral
Machines replacing people is concerning	negative
AI is boosting innovation across industries	positive
Job security is becoming uncertain due to AI	negative
AI might redefine how we work	neutral
AI is helping businesses grow faster	positive
Too much automation could harm workers	negative
AI is opening doors to new careers	positive
I think AI will balance jobs eventually	neutral
AI is reducing manual workload	positive
People are losing jobs because of automation	negative
AI impact is unpredictable right now	neutral
AI is enhancing productivity worldwide	positive
Automation is a threat to employment	negative
AI is the future of work	positive
No one knows how AI will change jobs completely	neutral
AI can replace repetitive tasks effectively	positive
Job losses due to AI are worrying	negative
AI might improve work-life balance	positive
I'm confused about AI's job impact	neutral
AI is helping companies scale quickly	positive
Automation could widen inequality	negative
AI is creating new opportunities daily	positive
AI changes are happening too fast	neutral
AI is revolutionizing industries	positive
Many workers fear AI takeover	negative
AI could lead to smarter workplaces	positive
The future of jobs is uncertain with AI	neutral
AI is beneficial for economic growth	positive
Automation might reduce job availability	negative
AI can improve efficiency significantly	positive
Hard to predict AI's long-term effects	neutral
AI is a powerful tool for development	positive

- Model Output Screenshot (precision, recall, F1-score results)

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Naive Bayes:
              precision    recall  f1-score   support

    negative       1.00      0.67      0.80         6
     neutral       0.67      0.40      0.50         5
    positive       0.69      1.00      0.82         9

    accuracy                           0.75        20
   macro avg       0.79      0.69      0.71        20
weighted avg       0.78      0.75      0.73        20
```

```
... Logistic Regression:
              precision    recall  f1-score   support

    negative       1.00      0.33      0.50         6
     neutral       0.33      0.20      0.25         5
    positive       0.60      1.00      0.75         9

    accuracy                           0.60        20
   macro avg       0.64      0.51      0.50        20
weighted avg       0.65      0.60      0.55        20
```

```
... SVM:
              precision    recall  f1-score   support

    negative       1.00      0.33      0.50         6
     neutral       0.33      0.20      0.25         5
    positive       0.60      1.00      0.75         9

    accuracy                           0.60        20
   macro avg       0.64      0.51      0.50        20
weighted avg       0.65      0.60      0.55        20
```

8. Limitations

- The dataset size is limited to only 100 tweets
- Data is manually created and may not fully represent real-world opinions
- Contextual nuances such as sarcasm and tone are difficult to capture
- External factors influencing opinions are not considered

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9. Future Improvements

- Collect real-time data using Twitter API
- Increase dataset size for better model performance
- Use advanced techniques such as TF-IDF or word embeddings
- Apply deep learning models like LSTM or BERT
- Perform hyperparameter tuning

10. AI Ethics & Responsible Usage Declaration

I hereby declare that:

- The dataset used in this assignment is properly created and ethically used
- No personal or sensitive data is included in this project
- The analysis has been conducted responsibly and for academic purposes only
- AI tools were used only for guidance and understanding

Dataset Details:

- Source: Manually created dataset inspired by real-world discussions
- Type of Data: Text (tweets)
- Contains Personal/Sensitive Data: No

Identified Bias:

The dataset may contain bias as it is manually created and may not fully represent diverse real-world opinions.

Responsible Usage Statement:

This project is intended solely for educational purposes. The results should not be used for real-world decision-making or professional applications.

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11. Student Details

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